HypArr

Computations with hyperplane arrangements

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Paul Mücksch

Paul Mücksch
Email: paul.muecksch+uni@gmail.com
Homepage: https://paulmuecksch.github.io
Address: Leibniz Universität Hannover
Institut für Algebra, Zahlentheorie und Diskrete Mathematik
Welfengarten 1
30167 Hannover, Germany

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Chapter 1

Introduction

HYPARR is a GAP 4 package for computations with hyperplane arrangements (see [OT92]), oriented matroids (see [BLVS+99]), their combinatorial and topological invariants, in particular geometric lattices, face posets, Milnor fibers and complements.

Chapter 2

Hyperplane arrangements

Let $\mathbb{K}$ be a field. A hyperplane arrangment is a finite set $\mathscr{A} = \{H_1, \ldots, H_n\}$ of hyperplanes in a finite dimensional $\mathbb{K}$-vector space V.

2.1 Construction of arrangements of hyperplanes

2.1.1 HyperplaneArrangement (for IsList)

▷ `HyperplaneArrangement(R)` (operation)

Returns: A hyperplane arrangement

Constructs a hyperplane arrangement from a list `R` of vectors representing defining linear forms of hyperplanes.

Each vector $[a_1, \ldots, a_\ell]$ in `R` corresponds to the hyperplane

$$a_1 x_1 + \cdots + a_\ell x_\ell = 0.$$

The vectors must lie in the same vector space.

Linearly dependent defining forms are removed automatically.

Example
```
gap> A := HyperplaneArrangement([[1,0,0],[0,1,0],[0,0,1]]);
<HyperplaneArrangement: 3 hyperplanes in 3-space>
gap> Roots(A);
[ [ 1, 0, 0 ], [ 0, 1, 0 ], [ 0, 0, 1 ] ]
```

2.2 Attributes of arrangements

2.2.1 Dimension (for IsHyperplaneArrangement)

▷ `Dimension(A)` (attribute)

Returns: A non-negative integer.

Returns the dimension of the ambient vector space in which the hyperplane arrangement `A` is defined.

Example
```
gap> A:=HyperplaneArrangement([[1,0,0],[0,1,0]]); Dimension(A);
<HyperplaneArrangement: 2 hyperplanes in 3-space>
3
```

2.2.2 Roots (for IsHyperplaneArrangement)

▷ `Roots(A)` (attribute)

Returns: A list of vectors.

Returns the list of vectors giving defining linear forms (also called *roots*) of the hyperplanes in the arrangement `A`.

Example

```
gap> A:=AGpql(3,3,2); Roots(A);
<HyperplaneArrangement: 3 hyperplanes in 2-space>
[ [ 1, -E(3) ], [ 1, -E(3)^2 ], [ 1, -1 ] ]
```

2.2.3 HArrDefField (for IsHyperplaneArrangement)

▷ `HArrDefField(A)` (attribute)

Returns: A field

Returns the field of definition of `A`. For further compuations, in characterisic 0 only cyclotomic fields are considered.

Example

```
gap> A:=HyperplaneArrangement(Z(3)^0*[[1,0],[0,1],[1,1]]); HArrDefField(A);
<HyperplaneArrangement: 3 hyperplanes in 2-space>
GF(3)
gap> A:=AGpql(5,5,2); HArrDefField(A);
<HyperplaneArrangement: 5 hyperplanes in 2-space>
CF(5)
```

2.2.4 IntersectionLattice (for IsHyperplaneArrangement)

▷ `IntersectionLattice(A)` (attribute)

Returns: An object of type `IsGeomLattice`.

Computes the intersection lattice of the hyperplane arrangement `A`.

The ground set `L` (see `LGroundSet` (2.4.1)) of the lattice is represented level-by-level. The entry `L[k]` contains all intersections of rank k.

Each lattice element is stored as a list of indices referring to hyperplanes in `Roots(A)`.

For example, the list `[1,3]` represents the intersection of the first and third hyperplane of the arrangement.

Example

```
gap> A := HyperplaneArrangement([[1,0,0],[0,1,0],[0,0,0],[1,1,1]]);
<HyperplaneArrangement: 3 hyperplanes in 3-space>
gap> L:=IntersectionLattice(A); LGroundSet(L);
<Geometric lattice: 3 atoms, rank 3>
[ [ [ 1 ], [ 2 ], [ 3 ] ],
  [ [ 1, 2 ], [ 1, 3 ], [ 2, 3 ] ],
  [ [ 1, 2, 3 ] ] ]
```

2.2.5 CharPoly (for IsHyperplaneArrangement)

▷ `CharPoly(A)` (attribute)

Returns: a univariate polynomial with integral coefficients.

Returns $\chi_{\mathscr{A}}$ the *characterisitc polynomial* of the arrangement `A`.

Example
```
gap> A:=AGpql(2,2,3);
<HyperplaneArrangement: 6 hyperplanes in 3-space>
gap> CharPoly(A);
t^3-6*t^2+11*t-6
```

2.2.6 ExpArr (for IsHyperplaneArrangement)

▷ `ExpArr(A)` (attribute)

Returns: a list of integers, or fail

Returns *exponents* of the arrangement $\mathscr{A}$ if the characteristic polynomial factors over the integers.

Example
```
gap> A:=HyperplaneArrangement([[1,0],[0,1],[1,1]]);
<HyperplaneArrangement: 3 hyperplanes in 2-space>
gap> ExpArr(A);
[ 1, 2 ]
```

2.2.7 MSetInvL (for IsHyperplaneArrangement)

▷ `MSetInvL(A)` (attribute)

Returns: An object of type IsGeomLattice.

Computes multiset invariants of the intersection lattice (see `IntersectionLattice` (2.2.4)) of the hyperplane arrangement `A`.

For each level of the lattice, the function records how many intersections occur with a given number of defining hyperplanes.

Example
```
gap> A:=HyperplaneArrangement([[1,0,0,],[0,1,0],[0,0,1],[1,1,1]]);
<HyperplaneArrangement: 4 hyperplanes in 3-space>
gap> MSetInvL(A);
[ [ [ 1, 4 ] ], [ [ 2, 6 ] ], [ [ 4, 1 ] ] ]
```

2.3 Properties of arrangements

2.3.1 IsReal (for IsHyperplaneArrangement)

▷ `IsReal(A)` (property)

Returns: true or false.

Determines, if the hyperplane arrangement is defined over the reals, i.e., if the entries of the defining linear forms are real.

2.3.2 HArrIsIrreducible (for IsHyperplaneArrangement)

▷ `HArrIsIrreducible(A)` (property)

Returns: true or false.

Determines, if the hyperplane arrangement is irreducible.

2.3.3 HArrIsGeneric (for IsHyperplaneArrangement)

▷ `HArrIsGeneric(A)` (property)

Returns: true or false.

Determines, if the hyperplane arrangement is generic. See also `LIsGeneric` (2.5.3).

Example
```
gap> A := HyperplaneArrangement([[1,0,0],[0,1,0],[0,0,1],[1,1,0],[1,1,1]]);
<HyperplaneArrangement: 5 hyperplanes in 3-space>
gap> HArrIsGeneric(A);
false
gap> A := HyperplaneArrangement([[1,0,0],[0,1,0],[0,0,1],[1,1,1]]);
<HyperplaneArrangement: 4 hyperplanes in 3-space>
gap> HArrIsGeneric(A);
true
```

2.4 Attributes of geometric lattices

2.4.1 LGroundSet (for IsGeomLattice)

▷ `LGroundSet(L)` (attribute)

Returns: list of lists

Returns the ground set of the geometric lattice `L`.

2.4.2 LAtoms (for IsGeomLattice)

▷ `LAtoms(L)` (attribute)

Returns: list

Returns the set of atoms of `L`.

2.4.3 LRank (for IsGeomLattice)

▷ `LRank(L)` (attribute)

Returns: a non-negative integer

Returns the rank of `L`.

2.4.4 LkFlats (for IsGeomLattice)

▷ `LkFlats(L)` (attribute)

Returns: a function

Returns a function extracting the flats of rank k in `L`.

2.4.5 LCircuits (for IsGeomLattice)

▷ `LCircuits(L)` (attribute)

Returns: a list of lists

Gives the circuits of `L`.

2.4.6 LRankFunction (for IsGeomLattice)

▷ `LRankFunction(L)` (attribute)

Returns: A function

Returns the rank function of the geometric lattice `L`.

2.4.7 LGraph (for IsGeomLattice)

▷ `LGraph(L)` (attribute)

Returns: a graph

Constructs the directed graph of the Hasse diagram of L.

2.4.8 LAutGroup (for IsGeomLattice)

▷ `LAutGroup(L)` (attribute)

Returns: a group

Computes the autmorphism group of L as a subgroup of `Sym(LAtoms(L))`.

2.4.9 LMoebius (for IsGeomLattice)

▷ `LMoebius(L)` (attribute)

Returns: function

Returns the Moebius function of `L`.

2.4.10 LCharPoly (for IsGeomLattice)

▷ `LCharPoly(L)` (attribute)

Returns: function

Computes the characteristic polynomial of `L`.

2.4.11 LBases (for IsGeomLattice)

▷ `LBases(L)` (attribute)

Returns: list of bases

Determines the subsets of `LAtoms(L)` which are bases.

Example

```
```

2.5 Properties of geometric lattices

2.5.1 LIsIrreducible (for IsGeomLattice)

▷ `LIsIrreducible(L)` (property)

Returns: true or false.

Determines, if L is irreducible.

2.5.2 LIsBoolean (for IsGeomLattice)

▷ LIsBoolean(*L*) (property)

Returns: true or false.

Determines, if L is a boolean lattice.

Example

```
gap> L:=IntersectionLattice(HyperplaneArrangement([[1,0],[1,1]]));
<Geometric lattice: 2 atoms, rank 2>
gap> LIsBoolean(L);
true
gap> L:=IntersectionLattice(HyperplaneArrangement([[1,0],[0,1],[1,1]]));
<Geometric lattice: 3 atoms, rank 2>
gap> LIsBoolean(L);
false
```

2.5.3 LIsGeneric (for IsGeomLattice)

▷ LIsGeneric(*L*) (property)

Returns: true or false.

Determines, if L is generic, i.e. if L is irreducible and all proper intervals are boolean.

Example

```
gap> A := HyperplaneArrangement([[1,0,0],[0,1,0],[0,0,1],[1,1,1]]);
<HyperplaneArrangement: 4 hyperplanes in 3-space>
gap> L:=IntersectionLattice(A);
<Geometric lattice: 4 atoms, rank 3>
gap> LIsGeneric(L);
true
gap> A := HyperplaneArrangement([[1,0,0],[0,1,0],[0,0,1],[1,1,0],[1,1,1]]);
<HyperplaneArrangement: 5 hyperplanes in 3-space>
gap> L:=IntersectionLattice(A);
<Geometric lattice: 5 atoms, rank 3>
gap> LIsGeneric(L);
false
```

2.6 Operations

2.6.1 LBsOfFlat (for IsGeomLattice, IsList)

▷ LBsOfFlat(*L*, *m*) (operation)

Returns: list of bases of m

Computes the bases of `m` in `L`.

Example

```
```

2.6.2 LBaseCircuit (for IsGeomLattice,IsList,IsInt)

▷ LBaseCircuit(*L*, *B*, *e*) (operation)

Returns: list

Determines the circuit contained in `Union(B,[e])`.

Example

```
```

2.6.3 LIsIsomorphic (for IsGeomLattice,IsGeomLattice)

▷ `LIsIsomorphic(LA, LB)` (operation)

Returns: true or false

Determines if the geometric lattices *LA* and *LB* are isomorphic.

Example
```
gap> A:=HyperplaneArrangement([[1,0],[0,1],[1,1]]); B:=HyperplaneArrangement([[1,0],[1,1],[1,2]]);
<HyperplaneArrangement: 3 hyperplanes in 2-space>
<HyperplaneArrangement: 3 hyperplanes in 2-space>
gap> LA:=IntersectionLattice(A); LB:=IntersectionLattice(B);
<Geometric lattice: 3 atoms, rank 2>
<Geometric lattice: 3 atoms, rank 2>
gap> LIsIsomorphic(LA,LB);
true
```

2.6.4 IsLEquiv (for IsHyperplaneArrangement,IsHyperplaneArrangement)

▷ `IsLEquiv(A, B)` (operation)

Returns: true or false

Determines if the arrangements *A* and *B* have isomorphic intersection lattices.

Example
```
gap> A:=HyperplaneArrangement(([[1,0,0],[0,1,0],[0,0,1],[1,1,0],[0,1,1],[1,1,3]]);
<HyperplaneArrangement: 6 hyperplanes in 3-space>
gap> B:=AGpql(2,2,3);
<HyperplaneArrangement: 6 hyperplanes in 3-space>
gap> IsLEquiv(A,B);
false
gap> A:=HyperplaneArrangement(([[1,0,0],[0,1,0],[0,0,1],[1,1,0],[0,1,1],[1,1,1]]);
<HyperplaneArrangement: 6 hyperplanes in 3-space>
gap> IsLEquiv(A,B);
true
```

2.6.5 Essentialization (for IsHyperplaneArrangement)

▷ `Essentialization(A)` (operation)

Returns: A hyperplane arrangement.

Computes the *essentialization* of the hyperplane arrangement *A*. An arrangement is called essential if the intersection of all its hyperplanes is the origin. If this is not the case, the function restricts the arrangement to a complementary subspace so that the resulting arrangement becomes essential.

Example
```
gap> A:=HyperplaneArrangement(([[1,0,0],[0,1,0],[1,1,0]]);
<HyperplaneArrangement: 3 hyperplanes in 3-space>
gap> B:=Essentialization(A); Roots(B);
<HyperplaneArrangement: 3 hyperplanes in 2-space>
[ [ 1, 0 ], [ 0, 1 ], [ 1, 1 ] ]
```

2.6.6 HArrRestriction (for IsHyperplaneArrangement, IsInt)

▷ `HArrRestriction(A, i)` (operation)

Returns: A hyperplane arrangement.

Computes the *restriction* of the arrangement `A` to the `i`-th hyperplane of `Roots(A)`.

2.6.7 HArrRestriction (for IsHyperplaneArrangement, IsList)

▷ `HArrRestriction(A, S)` (operation)

Returns: A hyperplane arrangement.

Computes the *restriction* of the arrangement `A` to the subspace orthogonal to the vectors in `S`.

2.6.8 HArrAddition (for IsHyperplaneArrangement, IsList)

▷ `HArrAddition(A, h)` (operation)

Returns: A hyperplane arrangement.

Computes the *addition* of the arrangement `A` with respect to hyperplane `h`.

2.6.9 HArrDeletion (for IsHyperplaneArrangement, IsInt)

▷ `HArrDeletion(A, i)` (operation)

Returns: A hyperplane arrangement.

Computes the *deletion* of the arrangement `A` with respect to `i`-th hyperplane of `Roots(A)`.

Chapter 3

Special arrangements

The following describes functions to construct some special arrangements.

3.1 Global methods

3.1.1 AGpql

▷ AGpql(*p*, *q*, *l*) (function)

Returns: A hyperplane arrangement.

Constructs the reflection arrangement associated to the monomial complex reflection group $G(p,q,l)$. The hyperplanes are defined by equations of the form

$$x_i = \zeta^k x_j$$

where ζ is a primitive p-th root of unity.

Example

```
gap> A := AGpql(2,1,3);
 <HyperplaneArrangement: 9 hyperplanes in 3-space>
gap> Roots(A);
[ [ 1, 0, 0 ], [ 0, 1, 0 ], [ 0, 0, 1 ],
  [ 1, 1, 0 ], [ 1, -1, 0 ], [ 1, 0, 1 ],
  [ 1, 0, -1 ], [ 0, 1, 1 ], [ 0, 1, -1 ] ]
```

3.1.2 SsS3

▷ SsS3(*n*) (function)

Returns: A hyperplane arrangement.

Generates the irreducible supersolvable simplicial arrangement of rank 3 with n hyperplanes. n must either be even oder congruent 1 mod 4. In any case, $n \geq 6$.

Example

```
gap> A:=SsS3(9); Roots(A);
<HyperplaneArrangement: 9 hyperplanes in 3-space>
[ [ 0, 0, 1 ], [ 0, 1, 0 ],
  [ -1/2*E(8)+1/2*E(8)^3, 1/2*E(8)-1/2*E(8)^3, 0 ],
  [ -1, 0, 0 ],
  [ -1/2*E(8)+1/2*E(8)^3, -1/2*E(8)+1/2*E(8)^3, 0 ],
```

```
  [ 1/2*E(8)-1/2*E(8)^3, 1/2*E(8)-1/2*E(8)^3, 1 ],
  [ -1/2*E(8)+1/2*E(8)^3, 1/2*E(8)-1/2*E(8)^3, 1 ],
  [ -1/2*E(8)+1/2*E(8)^3, -1/2*E(8)+1/2*E(8)^3, 1 ],
  [ 1/2*E(8)-1/2*E(8)^3, -1/2*E(8)+1/2*E(8)^3, 1 ] ]
```

3.1.3 ConnectedSubgraphArr (for IsList)

▷ `ConnectedSubgraphArr(Es)` (operation)

Returns: A hyperplane arrangement.

Generates the connected subgraph arrangement (see [CK22]) of the graph with edges `Es`.

Let $G = (N,E)$ be an undirected graph with vertex set $N = [n]$ and edge set E. For $I \subseteq N$, let $G[I]$ be the induced subgraph of G on the set of vertices I. For $I \subseteq N$, define the hyperplane

$$H_I := \ker \sum_{i \in I} x_i.$$

The connected subgraph arrangement consists of hyperplanes $\{H_I \mid \varnothing \neq I \subseteq N \text{ if } G[I] \text{ is connected } \}$.

Example

```
gap> A:=ConnectedSubgraphArr([[1,2],[1,3],[2,3]]); Roots(A);
<HyperplaneArrangement: 7 hyperplanes in 3-space>
[ [ 1, 1, 1 ], [ 0, 1, 1 ], [ 0, 0, 1 ], [ 0, 1, 0 ],
  [ 1, 0, 1 ], [ 1, 0, 0 ], [ 1, 1, 0 ] ]
```

3.1.4 GraphicArr (for IsList)

▷ `GraphicArr(Es)` (operation)

Returns: A hyperplane arrangement.

Generates the graphic arrangement of the graph with edges `Es`.

Let $G = (V,E)$ be an undirected graph with vertex set $V = [\ell]$ and edge set E. The graphic arrangement has hyperplanes

$$\{\{x_i = x_j\} \mid \{i,j\} \in E\}.$$

Example

Chapter 4

Oriented Matroids

Oriented matroids are abstractions of real hyperplane arrangements. The following describes functions to handle oriented matroids and compute associated cell complexes and invariants. See [BLVS+99].

4.1 Construction of oriented matroids

4.1.1 OrientedMatroid (for IsList)

▷ `OrientedMatroid(R, or, A, or[, CCs, c], or[, n, r, ChiroCoreList])` (operation)

Returns: An oriented matroid OM.

Constructs am oriented matroid from

- a list *R* of vectors representing defining linear forms of real hyperplanes,
- a real hyperplane arrangement *A*,
- a list of lists *CVs* of signed covectors and a string `c="cv"`,
- intergers *n*,*r* giving the size of the groundset and the rank and a list *ChiroCoreList* $\subseteq \{0,1,-1\}^m (m = \binom{n}{r})$ presenting the chirotope

of the oriented matroid.

Example

```
gap> A:=AGpql(2,2,3); OM:=OrientedMatroid(Roots(A));
<OrientedMatroid: 6 elements, rank 3>
gap> A:=AGpql(2,2,3); OM:=OrientedMatroid(A);
<OrientedMatroid: 6 elements, rank 3>
gap> OrientedMatroid([[[1,1],[-1,1],[1,-1],[-1,-1]],[[0,1],[0,-1],[1,0],[-1,0]],[[0,0]]],"cv");
<OrientedMatroid: 2 elements, rank 2>
gap> OrientedMatroid(2,3,[1,1,1]);
<OrientedMatroid: 3 elements, rank 2>
```

4.2 Attributes

4.2.1 OMRank (for IsOrientedMatroid)

▷ `OMRank(OM)` (attribute)

Returns: An integer

Returns the rank of the oriented matroid *OM*.

Example
```
gap> A := AGpql(2,2,3); OM := OrientedMatroid(A);
<OrientedMatroid: 6 elements, rank 3>
gap> OMRank(OM);
3
```

4.2.2 OMGroundSet (for IsOrientedMatroid)

▷ OMGroundSet(*OM*) (attribute)

Returns: A list

Returns the ground set of the oriented matroid OM.

Example
```
gap> A := AGpql(2,2,3); OM := OrientedMatroid(A);
<OrientedMatroid: 6 elements, rank 3>
gap> OMGroundSet(OM);
[ 1, 2, 3, 4, 5, 6 ]
```

4.2.3 OMLForms (for IsOrientedMatroid)

▷ OMLForms(*OM*) (attribute)

Returns: A list of linear forms or `fail`

Returns the list of linear forms associated with the oriented matroid OM. If the internal component `OM!.lforms` is not bound, `fail` is returned.

Example
```
gap> A := AGpql(2,2,3); OM := OrientedMatroid(A);
<OrientedMatroid: 6 elements, rank 3>
gap> OMLForms(O);
[ [ 1, 1, 0 ], [ 1, -1, 0 ], [ 1, 0, 1 ],
  [ 1, 0, -1 ], [ 0, 1, 1 ], [ 0, 1, -1 ] ]
```

4.2.4 OMGeomLattice (for IsOrientedMatroid)

▷ OMGeomLattice(*OM*) (attribute)

Returns: An object of type `IsGeomLattice` or `fail`

Constructs the geometric lattice associated with the oriented matroid OM. If OM is linear (as determined by `OMIsLinear(OM)`), the lattice is computed as the intersection lattice of the hyperplane arrangement given by `OMLForms(OM)`, stored in the internal component `OM!.lattice`, and returned.

Example
```
gap> O := OrientedMatroid([[1,0,0],[0,1,0],[0,0,1],[1,1,1]]);
<OrientedMatroid: 4 elements, rank 3>
gap> L:=OMGeomLattice(O); LGroundSet(L);
<Geometric lattice: 4 atoms, rank 3>
[ [ [ 1 ], [ 2 ], [ 3 ], [ 4 ] ],
  [ [ 1, 2 ], [ 1, 3 ], [ 2, 3 ],
      [ 1, 4 ], [ 2, 4 ], [ 3, 4 ] ],
  [ [ 1, 2, 3, 4 ] ] ]
gap> O := OrientedMatroid(3,4,[1,1,1,1]);
<OrientedMatroid: 4 elements, rank 3>
gap> L:=OMGeomLattice(O); LGroundSet(L);
```

```
<Geometric lattice: 4 atoms, rank 3>
[ [ [ 1 ], [ 2 ], [ 3 ], [ 4 ] ],
  [ [ 1, 2 ], [ 1, 3 ], [ 2, 3 ],
      [ 1, 4 ], [ 2, 4 ], [ 3, 4 ] ],
  [ [ 1, 2, 3, 4 ] ] ]
```

4.2.5 OMCocircuits (for IsOrientedMatroid)

▷ `OMCocircuits(OM)` (attribute)

Returns: A list

Computes the (signed) cocircuits of the oriented matroid OM.
The result is a list of sign vectors.

Example
```
gap> O:=OrientedMatroid([[1,0],[0,1],[1,1]]); OMCocircuits(O);
<OrientedMatroid: 3 elements, rank 2>
[ [ 0, 1, 1 ], [ 0, -1, -1 ], [ 1, 0, 1 ],
  [ -1, 0, -1 ], [ -1, 1, 0 ], [ 1, -1, 0 ] ]
```

4.2.6 OMCircuits (for IsOrientedMatroid)

▷ `OMCircuits(OM)` (attribute)

Returns: A list

Computes the (signed) circuits of the oriented matroid OM.
The result is a list of sign vectors.

Example
```
```

4.2.7 OMTopes (for IsOrientedMatroid)

▷ `OMTopes(OM)` (attribute)

Returns: A list

Returns the set of topes, i.e., the maximal covectors of *OM*

Example
```
gap> O := OrientedMatroid([[1,0],[0,1],[1,1]]);
<OrientedMatroid: 3 elements, rank 2>
gap> OMTopes(O);
[ [ -1, 1, -1 ], [ 1, -1, 1 ], [ -1, -1, -1 ],
    [ 1, 1, 1 ], [ -1, 1, 1 ], [ 1, -1, -1 ] ]
```

4.2.8 TopeGraph (for IsOrientedMatroid)

▷ `TopeGraph(OM)` (attribute)

Returns: A graph

Constructs the tope graph of the oriented matroid *OM*. The topes are obtained as the maximal covectors from `OMCovectors(OM)`. Two topes are connected by an edge if their separating set consists of one element.

Example
```
gap> O := OrientedMatroid([[1,0],[0,1],[1,1]]);
<OrientedMatroid: 3 elements, rank 2>
gap> G := ShallowCopy(TopeGraph(O));
gap> Vertices(G);
[ 1 .. 6 ]
gap> IsSimpleGraph(G);
true
gap> UndirectedEdges(G);
[ [ 1, 3 ], [ 1, 5 ], [ 2, 4 ], [ 2, 6 ], [ 3, 6 ], [ 4, 5 ] ]
```

4.2.9 OMCovectors (for IsOrientedMatroid)

▷ `OMCovectors(OM)` (attribute)

Returns: A list of lists

Computes all covectors of the oriented matroid OM. The result is returned as a list of lists, where the k-th inner list is a list of covectors corresponding to faces of dimension $k-1$.

Example
```
gap> O := OrientedMatroid([[1,0],[0,1],[1,1]]);
<OrientedMatroid: 3 elements, rank 2>
gap> OMCovectors(O);
[ [ [ -1, 1, -1 ], [ 1, -1, 1 ], [ -1, -1, -1 ],
    [ 1, 1, 1 ], [ -1, 1, 1 ], [ 1, -1, -1 ] ],
  [ [ 0, 1, 1 ], [ 0, -1, -1 ], [ 1, 0, 1 ],
    [ -1, 0, -1 ], [ -1, 1, 0 ], [ 1, -1, 0 ] ],
  [ [ 0, 0, 0 ] ] ]
```

4.2.10 OMChirotope (for IsOrientedMatroid)

▷ `OMChirotope(OM)` (attribute)

Returns: A function

Returns the chirotope given as a function from $\binom{E}{r} \to \{0, \pm 1\}$, where E is the groundset of OM and r is the rank of `OM`.

Example
```
gap> O:=OrientedMatroid(2,3,[1,1,1]); c:=OMChirotope(O);
<OrientedMatroid: 3 elements, rank 2>
function( S ) ... end
gap> List(Combinations([1..3],2),x->c(x));
[ 1, 1, 1 ]
```

4.2.11 OMRankFunction (for IsOrientedMatroid)

▷ `OMRankFunction(OM)` (attribute)

Returns: A function

Returns the rank function of the oriented matroid `OM`

4.3 Properties

4.3.1 OMIsLinear (for IsOrientedMatroid)

▷ `OMIsLinear(OM)` (property)

Returns: true or false

Determines whether the oriented matroid *OM* is linear. This is the case if the internal component `OM!.lforms` is bound.

Example
```
gap> A := AGpql(2,2,3); OM := OrientedMatroid(A);
<OrientedMatroid: 6 elements, rank 3>
gap> OMIsLinear(OM);
true
gap> O := OrientedMatroid(3,4,[1,1,1,1]);
<OrientedMatroid: 4 elements, rank 3>
gap> OMIsLinear(O);
false
```

4.3.2 HasSimpleSimplex (for IsOrientedMatroid)

▷ `HasSimpleSimplex(OM)` (property)

Returns: true or false

Returns whether the oriented matroid *OM* has a simple simplex, i.e. a simplicial Tope T such that the localizations at its vertices are boolean and there is another rank 1 covector not adjacent to T. If `Rank(OM)` ≥ 3 and this is true then its Salvetti complex can not be $K(\pi, 1)$.

Example
```
gap> O:=OrientedMatroid([[1,0,0,0],[0,1,0,0],[0,0,1,0],[0,0,0,1],[1,1,1,1]]);
<OrientedMatroid: 5 elements, rank 4>
gap> HasSimpleSimplex(O);
  - tope: [ -1, -1, -1, -1, -1 ]
  - vertices: [ [ 0, 0, 0, -1, -1 ], [ 0, 0, -1, 0, -1 ],
  [ 0, -1, 0, 0, -1 ], [ -1, 0, 0, 0, -1 ] ]
true
```

4.3.3 HasSimpleTriangle (for IsOrientedMatroid)

▷ `HasSimpleTriangle(OM)` (property)

Returns: true or false.

Checks all rank 3 localizations for a simple simplex, i.e. a simple triangle.

Example
```
```

4.4 Operations

4.4.1 IsLEquiv (for IsOrientedMatroid,IsOrientedMatroid)

▷ `IsLEquiv(OM1, OM2)` (operation)

Returns: true or false

Determines if the oriented matroids *OM1* and *OM2* have isomorphic geometric lattices.

Example

```
gap> O1:=OrientedMatroid(3,4,[1,1,1,1]);
<OrientedMatroid: 4 elements, rank 3>
gap> O2:=OrientedMatroid([[1,0,0],[0,1,0],[0,0,1],[1,1,1]]);
<OrientedMatroid: 4 elements, rank 3>
gap> IsLEquiv(O1,O2);
true
gap> O2:=OrientedMatroid([[1,0,0],[0,1,0],[0,0,1],[1,1,0]]);
<OrientedMatroid: 4 elements, rank 3>
gap> IsLEquiv(O1,O2);
false
```

4.4.2 HasSimpleSimplexRk (for IsOrientedMatroid, IsInt)

▷ `HasSimpleSimplexRk(OM, k)` (operation)

Returns: true or false

Returns whether the oriented matroid `OM` has a localiazion of rank `k` with a simple Simplex. If this is true for $k \geq 3$ then its Salvetti complex can not be $K(\pi, 1)$.

Example

```
gap> O:=OrientedMatroid([[1,0,0,0],[0,1,0,0],[0,0,1,0],[0,0,0,1],[1,1,1,0]]);
<OrientedMatroid: 5 elements, rank 4>
gap> HasSimpleSimplex(O);
false
gap> HasSimpleSimplexRk(O,3);
true
```

4.4.3 OMLocalizationRk (for IsOrientedMatroid, IsList, IsInt)

▷ `OMLocalizationRk(OM, F, k)` (operation)

Returns: An oriented matroid

Returns the rank `k` localization of `OM` at the flat `F`.

Example

```
gap> O:=OrientedMatroid([[1,0,0,0],[0,1,0,0],[0,0,1,0],[0,0,0,1],[1,1,1,0]]);
<OrientedMatroid: 5 elements, rank 4>
gap> L:=OMGeomLattice(O);
<Geometric lattice: 5 atoms, rank 4>
gap> LkFlats(L)(3);
[ [ 1, 2, 3, 5 ], [ 1, 2, 4 ], [ 1, 3, 4 ], [ 2, 3, 4 ], [ 1, 4, 5 ], [ 2, 4, 5 ], [ 3, 4, 5 ] ]
gap> OMLocalizationRk(O,[1,2,3,5],3);
<OrientedMatroid: 4 elements, rank 3>
```

4.4.4 OMTConvexClosure (for IsOrientedMatroid, IsList)

▷ `OMTConvexClosure(OM, Ts)` (operation)

Returns: A list of topes (sign vectors)

Computes the convex closure of the set of topes `Ts` in `OM`.

Example

4.5 Global methods

4.5.1 OMOperation

▷ `OMOperation(sv1, sv2)` (function)

Returns: A list

Performs the oriented matroid operation on two signed vectors `sv1` and `sv2`. For each position `i`, if `sv1[i]` is nonzero, it is used; otherwise `sv2[i]` is used. Returns a new signed vector representing the result of the operation.

Example

```
gap> CVs:=OMCovectors(O)[2];
[ [ 0, 1, 1 ], [ 0, -1, -1 ], [ 1, 0, 1 ],
  [ -1, 0, -1 ], [ -1, 1, 0 ], [ 1, -1, 0 ] ]
gap> sv1 := CVs[3]; sv2 := CVs[5];
[ 1, 0, 1 ]
[ -1, 1, 0 ]
gap> OMOperation(sv1, sv2);
[ 1, 1, 1 ]
gap> OMOperation(sv2, sv1);
[ -1, 1, 1 ]
```

4.5.2 OrderCovec

▷ `OrderCovec(arg)` (function)

Order function for sign vectors.

4.5.3 IsOMEquiv

▷ `IsOMEquiv(OM1, OM2)` (function)

Returns: true or false

Determines, if the oriented matroids *OM1* and *OM2* are isomorphic.

4.5.4 SeparatingSet

▷ `SeparatingSet(arg)` (function)

The separating set of two sign vectors.

Chapter 5

Free arrangements

The following describes functions to analyse freeness properties of arrangements such as *inductive freeness* due to H. Terao [Ter80] and *divisionell freeness* due to T. Abe [Abe16].

5.1 Module of logarithmic vector fields aka derivation modules of arrangements

The functions in this section all use the SINGULAR package to call functions from SINGULAR [DGPS24] for commutative algebra calculations.

5.1.1 DerModule (for IsHyperplaneArrangement)

▷ `DerModule(A[, mult])` (operation)

Returns: Derivation module

Computes the *(multi) derivation module* of the arrangement `A` (with multiplicities given as a list by `mult`).

Example

```
gap> A:=HyperplaneArrangement([[1,0,0],[0,1,0],[0,0,1],[1,1,1]]);
<HyperplaneArrangement: 4 hyperplanes in 3-space>
gap> D:=DerModule(A);
<Derivation module
  over PolynomialRing( Rationals, ["x_1", "x_2", "x_3"] )
  with 4 generators>
gap> D:=DerModule(A,[2,1,3,4]);
<Derivation module
  over PolynomialRing( Rationals, ["x_1", "x_2", "x_3"] )
  with 4 generators>
```

5.1.2 DerPModule (for IsHyperplaneArrangement, IsInt)

▷ `DerPModule(A, p[, mult])` (operation)

Returns: Derivation module

Computes the *(multi) p-derivation module* $D^p(A)$ ($D^p(A,m)$) of the arrangement `A` (with multiplicities given as a list by `mult`).

Example

5.1.3 DerModGenerators (for IsDerivationModule)

▷ DerModGenerators(*D*) (attribute)

Returns: list

Returns the (minimal) generators of the module *D*.

Example
```
gap> A:=HyperplaneArrangement([[1,0,0],[0,1,0],[0,0,1],[1,1,1]]);
<HyperplaneArrangement: 4 hyperplanes in 3-space>
gap> D:=DerModule(A);
<Derivation module
  over PolynomialRing( Rationals, ["x_1", "x_2", "x_3"] )
  with 4 generators>
gap> DerModGenerators(D);
[ [ x_1, x_2, x_3 ],
  [ 0, -x_2*x_3, x_2*x_3 ],
  [ 0, x_2*x_3, x_1*x_3+x_3^2 ],
  [ 0, x_1*x_2+x_2^2+x_2*x_3, 0 ] ]
```

5.1.4 DerModPRing (for IsDerivationModule)

▷ DerModPRing(*D*) (attribute)

Returns: Polynomial ring

Returns the defining polynomial ring of *D*.

Example
```
gap> D:=DerModule(AGpql(3,3,2)); DerModPRing(D);
<field in characteristic 0>[x_1,x_2]
```

5.1.5 DerModDegreeSequence (for IsDerivationModule)

▷ DerModDegreeSequence(*D*) (attribute)

Returns: list

Returns the sequence of degrees of (minimal) generators of the graded module *D*.

Example
```
gap> A:=HyperplaneArrangement([[1,0,0],[0,1,0],[0,0,1],[1,1,0],[1,1,1]]);
<HyperplaneArrangement: 5 hyperplanes in 3-space>
gap> DerModDegreeSequence(DerModule(A));
[ 1, 2, 2 ]
```

5.1.6 DerModProjDim (for IsDerivationModule)

▷ DerModProjDim(*D*) (attribute)

Returns: A non-negative integer

Computes the projective dimension of the module *D*.

Example
```
gap> A:=HyperplaneArrangement([[1,0,0],[0,1,0],[0,0,1],[1,1,1]]);
<HyperplaneArrangement: 4 hyperplanes in 3-space>
gap> D:=DerModule(A);
<Derivation module
  over PolynomialRing( Rationals, ["x_1", "x_2", "x_3"] )
  with 4 generators>
```

```
gap> DerModProjDim(D);
1
```

5.2 Freeness properties

5.2.1 DerModIsFree (for IsDerivationModule)

▷ `DerModIsFree(D)` (property)

Returns: true or false

Determines, if D is a free module over `DerModPRing(D)`.

Example

```
gap> A:=HyperplaneArrangement([[1,0,0],[0,1,0],[0,0,1],[1,1,0],[1,1,1]]);
<HyperplaneArrangement: 5 hyperplanes in 3-space>
gap> D:=DerModule(A); DerModIsFree(D);
true
gap> A:=HyperplaneArrangement([[1,0,0],[0,1,0],[0,0,1],[1,1,1]]);
<HyperplaneArrangement: 4 hyperplanes in 3-space>
gap> D:=DerModule(A); DerModIsFree(D);
false
```

5.2.2 HArrIsFree (for IsHyperplaneArrangement)

▷ `HArrIsFree(A)` (property)

Returns: true or false

Determines, if A is free, i.e. if the derivation module is a free module over the coordinate ring of the ambient space.

Example

```
gap> A:=HyperplaneArrangement([[1,0,0],[0,1,0],[0,0,1],[1,1,0],[1,1,1]]);
<HyperplaneArrangement: 5 hyperplanes in 3-space>
gap> HArrIsFree(A);
true
gap> A:=HyperplaneArrangement([[1,0,0],[0,1,0],[0,0,1],[1,1,1]]);
<HyperplaneArrangement: 4 hyperplanes in 3-space>
gap> HArrIsFree(A);
false
```

5.2.3 HArrIsLocallyFree (for IsHyperplaneArrangement)

▷ `HArrIsLocallyFree(A)` (property)

Returns: true or false

Determines, if A is locally free, i.e. if all proper localizations are free.

Example

```
```

5.2.4 IsInductivelyFree (for IsHyperplaneArrangement)

▷ `IsInductivelyFree(A)` (property)

Returns: true or false.

Determines whether the arrangement A is *inductively free*.

Example

```
gap> A:=HyperplaneArrangement([[1,0,0],[0,1,0],[0,0,1],[1,1,0],[1,1,1]]);
<HyperplaneArrangement: 5 hyperplanes in 3-space>
gap> IsInductivelyFree(A);
true
```

5.2.5 IsLocallyInductivelyFree (for IsHyperplaneArrangement)

▷ `IsLocallyInductivelyFree(A)` (property)

Returns: true or false

Determines, if `A` is locally free, i.e. if all proper localizations are inductively free.

Example

```
```

5.2.6 IsDivisionallyFree (for IsHyperplaneArrangement)

▷ `IsDivisionallyFree(A)` (property)

Returns: true or false.

Determines whether the arrangement `A` is *divisionally free*.

Example

```
gap> A:=HyperplaneArrangement([[1,0,0],[0,1,0],[0,0,1],[1,1,0],[1,1,1]]);
<HyperplaneArrangement: 5 hyperplanes in 3-space>
gap> IsDivisionallyFree(A);
true
gap> A:=AGpql(3,3,3);
<HyperplaneArrangement: 9 hyperplanes in 3-space>
gap> IsDivisionallyFree(A);
false
```

5.2.7 IsLocallyDivisionallyFree (for IsHyperplaneArrangement)

▷ `IsLocallyDivisionallyFree(A)` (property)

Returns: true or false

Determines, if `A` is locally free, i.e. if all proper localizations are divisionallys free.

Example

```
```

Chapter 6

Further properties of arrangements, geometric lattices, and oriented matroids

The following describes functions to analyse further properties of arrangements such as *formality*, *supersolvability*, *simpliciality*, *factoredness*, *inductive factoredness* ...

6.1 Supersolvable arrangements

6.1.1 LIsModularPair (for IsGeomLattice, IsList, IsList)

▷ `LIsModularPair(L, m1, m2)` (operation)

Returns: true or false

Determines if *m1* and *m2* form a modular pair in L.

Example
```
gap> A:=AGpql(2,2,3);
<HyperplaneArrangement: 6 hyperplanes in 3-space>
gap> L:=IntersectionLattice(A);
<Geometric lattice: 6 atoms, rank 3>
gap> LGroundSet(L);
[ [ [ 1 ], [ 2 ], [ 3 ], [ 4 ], [ 5 ], [ 6 ] ],
  [ [ 1, 2 ], [ 1, 3, 6 ], [ 2, 3, 5 ], [ 1, 4, 5 ],
    [ 2, 4, 6 ], [ 3, 4 ], [ 5, 6 ] ],
  [ [ 1, 2, 3, 4, 5, 6 ] ] ]
gap> m1:=[1,2]; m2:= [5,6]; LIsModularPair(L,m1,m2);
false
gap> m1:=[1,2]; m2:= [2,4,6]; LIsModularPair(L,m1,m2);
true
```

6.1.2 LIsModularFlat (for IsGeomLattice, IsList)

▷ `LIsModularFlat(L, m)` (operation)

Returns: true or false

Determines if *m* is a modular flat in L.

Example
```
gap> A:=AGpql(2,2,3); L:=IntersectionLattice(A); LGroundSet(L);
<HyperplaneArrangement: 6 hyperplanes in 3-space>
```

```
<Geometric lattice: 6 atoms, rank 3>
[ [ [ 1 ], [ 2 ], [ 3 ], [ 4 ], [ 5 ], [ 6 ] ],
  [ [ 1, 2 ], [ 1, 3, 6 ], [ 2, 3, 5 ], [ 1, 4, 5 ],
    [ 2, 4, 6 ], [ 3, 4 ], [ 5, 6 ] ],
  [ [ 1, 2, 3, 4, 5, 6 ] ] ]
gap> m1:=[1,2]; m2:= [2,4,6]; LIsModularFlat(L,m1); LIsModularFlat(L,m2);
false
true
```

6.1.3 LModularFlatsRk (for IsGeomLattice, IsInt)

▷ LModularFlatsRk(*L*, *k*) (operation)

Returns: list

Determines the modular flats in *L* of rank *k*.

Example
```
gap> L:=IntersectionLattice(AGpql(2,1,4));
<Geometric lattice: 16 atoms, rank 4>
gap> LModularFlatsRk(L,3);
[ [ 1, 2, 3, 5, 6, 7, 8, 11, 12 ],
  [ 1, 2, 4, 5, 6, 9, 10, 13, 14 ],
  [ 1, 3, 4, 7, 8, 9, 10, 15, 16 ],
  [ 2, 3, 4, 11, 12, 13, 14, 15, 16 ] ]
```

6.1.4 LIsSupersolvable (for IsGeomLattice)

▷ LIsSupersolvable(*L*) (property)

Returns: true or false.

Determines whether the geometric lattice *L* is *supersolvable*.

Example
```
gap> L:=IntersectionLattice(AGpql(2,1,4));
<Geometric lattice: 16 atoms, rank 4>
gap> LIsSupersolvable(L);
true
gap> L:=IntersectionLattice(AGpql(2,2,4));
<Geometric lattice: 12 atoms, rank 4>
gap> LIsSupersolvable(L);
false
```

6.1.5 HArrIsSupersolvable (for IsHyperplaneArrangement)

▷ HArrIsSupersolvable(*A*) (property)

Returns: true or false.

Determines whether the arrangement *A* is *supersolvable*.

Example
```
gap> A:=AGpql(2,1,4);
<HyperplaneArrangement: 16 hyperplanes in 4-space>
gap> HArrIsSupersolvable(A);
true
gap> A:=AGpql(2,2,4);
<HyperplaneArrangement: 12 hyperplanes in 4-space>
```

```
gap> HArrIsSupersolvable(A);
false
```

6.1.6 OMIsSupersolvable (for IsOrientedMatroid)

▷ OMIsSupersolvable(*OM*) (property)

Returns: true or false.

Determines whether the oriented matroid *OM* is *supersolvable*.

Example
```
gap> O:=OrientedMatroid(3,4,[1,1,1,1]);
<OrientedMatroid: 4 elements, rank 3>
gap> OMIsSupersolvable(O);
false
gap> C:=List(Combinations(Roots(AGpql(2,2,3)),3),x->pos(Determinant(x)));
[ 1, 1, 1, 1, 0, -1, -1, -1, -1, 0, 1, 0, 1, -1, 0, 1, 1, 1, -1, -1 ]
gap> O:=OrientedMatroid(3,6,C);
<OrientedMatroid: 6 elements, rank 3>
gap> OMIsSupersolvable(O);
true
```

6.2 Formal arrangements

6.2.1 IsFormal (for IsHyperplaneArrangement)

▷ IsFormal(*A*) (property)

Returns: true or false.

Determines whether the arrangement is *formal* in the sense of Falk-Randell [FR87].

Example
```
gap> A1 := Arr([
>     [1,0,0],
>     [0,1,0],
>     [0,0,1],
>     [1,1,1],
>     [2,1,1],
>     [2,3,1],
>     [2,3,4],
>     [3,0,5],
>     [3,4,5]
> ]);
gap>
gap> A2 := Arr([
>     [1,0,0],
>     [0,1,0],
>     [0,0,1],
>     [1,1,1],
>     [2,1,1],
>     [2,3,1],
>     [2,3,4],
>     [1,0,3],
>     [1,2,3]
```

```
> ]);
gap> IsLEquiv(A1,A2);
true
gap> IsFormal(A1);
true
gap> IsFormal(A2);
false
```

6.3 Simplicial arrangements

6.3.1 HArrIsSimplicial (for IsHyperplaneArrangement and IsReal)

▷ `HArrIsSimplicial(A)` (property)

Returns: true or false.

Determines whether the real arrangement `A` is *simplicial*, i.e. if all chambers are simplicial.

Example

```
gap> A:=HyperplaneArrangement([[1,0,0],[0,1,0],[0,0,1],[1,1,0],[0,1,1]]); HArrIsSimplicial(A);
<HyperplaneArrangement: 5 hyperplanes in 3-space>
false
gap> A:=HyperplaneArrangement([[1,0,0],[0,1,0],[0,0,1],[1,1,0],[0,1,1],[1,1,1]]); HArrIsSimplicia
<HyperplaneArrangement: 6 hyperplanes in 3-space>
true
```

6.4 Falk's weight test

The following still needs to be tested further...

6.4.1 OMBoundedCpx (for IsOrientedMatroid, IsInt)

▷ `OMBoundedCpx(OM, g)` (operation)

Returns: list of list of sign vectors.

Constructs the complex of bounded cells of the affine part of the oriented matroid `OM` with respect to the element g.

Example

```
```

6.4.2 OMSupportsFalkWeights (for IsOrientedMatroid, IsInt)

▷ `OMSupportsFalkWeights(OM, g)` (operation)

Returns: list or fail

Only for oriented matroids of rank 3. Determines, whether `OM` supports a system of weights as described by M. Falk in [Fal95]. This implies, that the Salvetti complex (see `SalvettiComplex` (7.1.1)) is apsherical. The functionality of the CDDINTERFACE package is used, to determine solutions of a system of linear inequalities.

If the algorithm determines a suitable weight system for `OM`, each list element consists of a weight, and a corner, i.e. a pair of a 2-cell and an adjacent vertex in the bounded complex, given as sign vectors.

Example
```
gap> O:=OrientedMatroid(
>       [[1,1,0],[1,-1,0],[1,0,1],[1,0,-1],[0,1,1],[0,1,-1],[0,0,1]]
> );
<OrientedMatroid: 7 elements, rank 3>
gap> OMIsSupersolvable(O);
false
gap> OMSupportsFalkWeights(O,3);
fail
gap> OMSupportsFalkWeights(O,1);
[ [ 1/2, [ [ 1, 1, 1, 1, 1, 1, 1 ], [ 1, 0, 1, 0, 1, 0, 1 ] ] ],
  [ 0, [ [ 1, 1, 1, 1, 1, 1, 1 ], [ 1, 1, 1, 1, 0, 0, 0 ] ] ],
  [ 1/2, [ [ 1, 1, 1, 1, 1, 1, 1 ], [ 1, 0, 1, 1, 1, 1, 0 ] ] ],
  [ 0, [ [ 1, 1, 1, 1, 1, 1, -1 ], [ 1, 0, 0, 1, 0, 1, -1 ] ] ],
  [ 1/2, [ [ 1, 1, 1, 1, 1, 1, -1 ], [ 1, 1, 1, 1, 0, 0, 0 ] ] ],
  [ 1/2, [ [ 1, 1, 1, 1, 1, 1, -1 ], [ 1, 0, 1, 1, 1, 1, 0 ] ] ],
  [ 0, [ [ 1, -1, 1, 1, 1, 1, 1 ], [ 1, 0, 1, 0, 1, 0, 1 ] ] ],
  [ 1/2, [ [ 1, -1, 1, 1, 1, 1, 1 ], [ 1, -1, 0, 0, 1, 1, 0 ] ] ],
  [ 1/2, [ [ 1, -1, 1, 1, 1, 1, 1 ], [ 1, 0, 1, 1, 1, 1, 0 ] ] ],
  [ 1/2, [ [ 1, -1, 1, 1, 1, 1, -1 ], [ 1, 0, 0, 1, 0, 1, -1 ] ] ],
  [ 0, [ [ 1, -1, 1, 1, 1, 1, -1 ], [ 1, -1, 0, 0, 1, 1, 0 ] ] ],
  [ 1/2, [ [ 1, -1, 1, 1, 1, 1, -1 ], [ 1, 0, 1, 1, 1, 1, 0 ] ] ] ]
```

6.5 Factored arrangements

6.5.1 HArrIsFactored (for IsHyperplaneArrangement)

▷ `HArrIsFactored(A)` (attribute)

Returns: A list or fail

Determines if A is factored and if so returns some factorization.

Example
```
gap> A:=AGpql(2,1,3);
<HyperplaneArrangement: 9 hyperplanes in 3-space>
gap> HArrIsFactored(A);
[ [ 1 ], [ 2, 4, 5 ], [ 3, 6, 7, 8, 9 ] ]
gap> A:=AGpql(2,2,4);
<HyperplaneArrangement: 12 hyperplanes in 4-space>
gap> HArrIsFactored(A);
fail
```

6.5.2 HArrFactorizations (for IsHyperplaneArrangement)

▷ `HArrFactorizations(A)` (operation)

Returns: A list

Computes a list of all factorizations of A as partitions of $[1..|A|]$.

Example
```
gap> A:=AGpql(2,1,3);
<HyperplaneArrangement: 9 hyperplanes in 3-space>
gap> HArrFactorizations(A);
[ [ [ 1 ], [ 2, 4, 5 ], [ 3, 6, 7, 8, 9 ] ],
```

```
  [ [ 1 ], [ 2, 4, 5, 8, 9 ], [ 3, 6, 7 ] ],
  [ [ 2 ], [ 1, 4, 5 ], [ 3, 6, 7, 8, 9 ] ],
  [ [ 2 ], [ 1, 4, 5, 6, 7 ], [ 3, 8, 9 ] ],
  [ [ 3 ], [ 1, 6, 7 ], [ 2, 4, 5, 8, 9 ] ],
  [ [ 3 ], [ 1, 4, 5, 6, 7 ], [ 2, 8, 9 ] ],
  [ [ 4 ], [ 1, 2, 5 ], [ 3, 6, 7, 8, 9 ] ],
  [ [ 5 ], [ 1, 2, 4 ], [ 3, 6, 7, 8, 9 ] ],
  [ [ 6 ], [ 1, 3, 7 ], [ 2, 4, 5, 8, 9 ] ],
  [ [ 7 ], [ 1, 3, 6 ], [ 2, 4, 5, 8, 9 ] ],
  [ [ 8 ], [ 1, 4, 5, 6, 7 ], [ 2, 3, 9 ] ],
  [ [ 9 ], [ 1, 4, 5, 6, 7 ], [ 2, 3, 8 ] ] ]
```

6.5.3 HArrIsInductivelyFactored (for IsHyperplaneArrangement)

▷ HArrIsInductivelyFactored(*A*) (attribute)

Returns: A list or fail

Determines if *A* is inductively factored and if so returns some inductive factorization.

Example

```
gap> A:=AGpql(2,1,4);
<HyperplaneArrangement: 16 hyperplanes in 4-space>
gap> HArrIsInductivelyFactored(A);
[ [ [ 1 ], [ 2, 5, 6 ], [ 3, 7, 8, 11, 12 ], [ 4, 9, 10, 13, 14, 15, 16 ] ], true ]
gap> A:=AGpql(2,2,4);
<HyperplaneArrangement: 12 hyperplanes in 4-space>
gap> HArrIsInductivelyFactored(A);
false
```

6.5.4 HArrInductiveFactorizations (for IsHyperplaneArrangement)

▷ HArrInductiveFactorizations(*A*) (operation)

Returns: A list

Computes a list of all inductive factorizations of *A* as partitions of $[1..|A|]$.

Example

```
gap> A:=AGpql(2,1,3);
<HyperplaneArrangement: 9 hyperplanes in 3-space>
gap> HArrInductiveFactorizations(A);
[ [ [ 1 ], [ 2, 4, 5 ], [ 3, 6, 7, 8, 9 ] ],
  [ [ 1 ], [ 2, 4, 5, 8, 9 ], [ 3, 6, 7 ] ],
  [ [ 2 ], [ 1, 4, 5 ], [ 3, 6, 7, 8, 9 ] ],
  [ [ 2 ], [ 1, 4, 5, 6, 7 ], [ 3, 8, 9 ] ],
  [ [ 3 ], [ 1, 6, 7 ], [ 2, 4, 5, 8, 9 ] ],
  [ [ 3 ], [ 1, 4, 5, 6, 7 ], [ 2, 8, 9 ] ],
  [ [ 4 ], [ 1, 2, 5 ], [ 3, 6, 7, 8, 9 ] ],
  [ [ 5 ], [ 1, 2, 4 ], [ 3, 6, 7, 8, 9 ] ],
  [ [ 6 ], [ 1, 3, 7 ], [ 2, 4, 5, 8, 9 ] ],
  [ [ 7 ], [ 1, 3, 6 ], [ 2, 4, 5, 8, 9 ] ],
  [ [ 8 ], [ 1, 4, 5, 6, 7 ], [ 2, 3, 9 ] ],
  [ [ 9 ], [ 1, 4, 5, 6, 7 ], [ 2, 3, 8 ] ] ]
```

Chapter 7

Complements of complex arrangements

For a hyperplane arrangement $\mathscr{A}$ in $\mathbb{C}^\ell$, the following provides functions to compute complexes which have the homotopy type of the complement manifold $\mathbb{C}^\ell \setminus \bigcup_{H\in\mathscr{A}} H$.

7.1 Complexes and face posets

7.1.1 SalvettiComplex (for IsOrientedMatroid)

▷ SalvettiComplex(*OM, or, A*) (attribute)

Returns: A FacePoset

Constructs the Salvetti complex [Sal87] associated with the oriented matroid OM or the real hyperplane arrangement *A*.

Example
```
gap> O := OrientedMatroid([[1,0],[0,1],[1,1]]);
<OrientedMatroid: 3 elements, rank 2>
gap> SalvettiComplex(O);
<FacePoset of dimension 2 with f-vector [ 6, 12, 6 ]>
gap> A:=HyperplaneArrangement([[1,0],[0,1],[1,1]]);
<HyperplaneArrangement: 3 hyperplanes in 2-space>
gap> SalvettiComplex(A);
<FacePoset of dimension 2 with f-vector [ 6, 12, 6 ]>
```

7.1.2 SalvettiComplex (for IsHyperplaneArrangement)

▷ SalvettiComplex(*A*) (attribute)

See SalvettiComplex (7.1.1).

7.1.3 FPGroundSet (for IsFacePoset)

▷ FPGroundSet(*FP*) (attribute)

Returns: A list

Returns the ground set of the face poset FP.

Example
```
gap> S := SalvettiComplex(O);
<FacePoset of dimension 2 with f-vector [ 6, 12, 6 ]>
```

```
gap> FPGroundSet(S);
[
  [
    [ [ -1, 1, -1 ], [ -1, 1, -1 ] ],
    [ [ 1, -1, 1 ], [ 1, -1, 1 ] ],
    [ [ -1, -1, -1 ], [ -1, -1, -1 ] ],
    [ [ 1, 1, 1 ], [ 1, 1, 1 ] ],
    [ [ -1, 1, 1 ], [ -1, 1, 1 ] ],
    [ [ 1, -1, -1 ], [ 1, -1, -1 ] ]
  ],
  [
    [ [ -1, 0, -1 ], [ -1, 1, -1 ] ],
    [ [ -1, 1, 0 ], [ -1, 1, -1 ] ],
    [ [ 1, 0, 1 ], [ 1, -1, 1 ] ],
    [ [ 1, -1, 0 ], [ 1, -1, 1 ] ],
    [ [ 0, -1, -1 ], [ -1, -1, -1 ] ],
    [ [ -1, 0, -1 ], [ -1, -1, -1 ] ],
    [ [ 0, 1, 1 ], [ 1, 1, 1 ] ],
    [ [ 1, 0, 1 ], [ 1, 1, 1 ] ],
    [ [ 0, 1, 1 ], [ -1, 1, 1 ] ],
    [ [ -1, 1, 0 ], [ -1, 1, 1 ] ],
    [ [ 0, -1, -1 ], [ 1, -1, -1 ] ],
    [ [ 1, -1, 0 ], [ 1, -1, -1 ] ]
  ],
  [
    [ [ 0, 0, 0 ], [ -1, 1, -1 ] ],
    [ [ 0, 0, 0 ], [ 1, -1, 1 ] ],
    [ [ 0, 0, 0 ], [ -1, -1, -1 ] ],
    [ [ 0, 0, 0 ], [ 1, 1, 1 ] ],
    [ [ 0, 0, 0 ], [ -1, 1, 1 ] ],
    [ [ 0, 0, 0 ], [ 1, -1, -1 ] ]
  ]
]
```

7.1.4 FPOrder (for IsFacePoset)

▷ `FPOrder(FP)` (attribute)

Returns: A function

Returns the order function of the face poset FP.

7.1.5 BZs1Complex (for IsHyperplaneArrangement)

▷ `BZs1Complex(A)` (attribute)

Returns: A `FacePoset`

Constructs the complex described by Bjoener and Ziegler in [BZ92] oibtained from the $s^{(1)}$-stratification of a complex hyperplane arrangement A which has the homotopy type of the complement.

Example

```
gap> A:=AGpql(3,3,3);
<HyperplaneArrangement: 9 hyperplanes in 3-space>
gap> Cs1:=BZs1Complex(A);
```

```
<FacePoset of dimension 4 with f-vector [ 194, 1004, 1560, 812, 62 ]>
gap> Cs1CW := FPtoCWCpx(Cs1);
Regular CW-complex of dimension 4
gap> List([0..4],x->Length(Homology(Cs1CW,x)));
[ 1, 9, 24, 16, 0 ]
gap> CharPoly(A);
t^3-9*t^2+24*t-16
```

7.1.6 BZs2Complex (for IsHyperplaneArrangement)

▷ `BZs2Complex(A)` (attribute)

Returns: A `FacePoset`

Constructs the complex described by Bjoener and Ziegler in [BZ92] oibtained from the $s^{(2)}$-stratification of a complex hyperplane arrangement `A` which has the homotopy type of the complement.

Example
```
gap> A:=AGpql(3,3,2);
<HyperplaneArrangement: 3 hyperplanes in 2-space>
gap> Cs2:=BZs2Complex(A);
<FacePoset of dimension 4 with f-vector [ 52, 132, 96, 16, 0 ]>
```

Be aware that both functions `BZs1Complex` and `BZs2Complex` may take a considerable amount of time computing the complex for non-real arrangements since oriented matroid complexes in double dimension and double number of hyperplanes need to be computed. For a real arrangement, the $s^{(1)}$-complex is isomorphic to the Salvetti complex.

7.2 Non-$K(\pi,1)$ arrangements

7.2.1 CCSimpleTriangle (for IsHyperplaneArrangement)

▷ `CCSimpleTriangle(A)` (operation)

Returns: true or false

Tests if there is a convex open subset of the ambient space, such that the arrangement restricted to this convex open subset is a "simple triangle", see also `HasSimpleTriangle` (4.3.3). This implies that the arrangement is not $K(\pi,1)$.

Example
```
gap> A:=Arr([ [ E(4), E(4), E(4) ], [ 0, E(4), 1 ], [ 1, 0, 1 ], [ 1, 0, 0 ] ]);
<HyperplaneArrangement: 4 hyperplanes in 3-space>
gap> CCSimpleTriangle(A);
[ 1, 2, 3 ]
true
```

Chapter 8

Milnor fibers

The Milnor fiber is the typical fiber $f^{-1}(1)$ of the Milnor fibration

$$f : \mathbb{C}^\ell \setminus \cup_{H\in\mathscr{A}} H \to \mathbb{C}^\times, z \mapsto \prod_{H\in\mathscr{A}} \alpha_H(z)$$

of the hyperplane arrangement $\mathscr{A}$, where α_H are defining linear forms.

The following describes functions to construct a regular cell complex having the homotopy type of the Milnor fiber from [MY25] and thus enables computation of homotopy invaraints using e.g. the HAP package [Ell26].

8.1 Complexes

8.1.1 MilnorFiberComplex (for IsOrientedMatroid)

▷ `MilnorFiberComplex(OM)` (attribute)

Returns: FacePoset

Computes the face poset of the regular cell complex of the combinatorial Milnor fiber of the oriented matroid *OM*.

Example
```
gap> O:=OrientedMatroid([[1,0],[0,1],[1,1]]);
<OrientedMatroid: 3 elements, rank 2>
gap> MCpx:=MilnorFiberComplex(O);
<FacePoset of dimension 1 with f-vector [ 3, 6 ]>
```

8.1.2 MilnorFiberComplex (for IsHyperplaneArrangement)

▷ `MilnorFiberComplex(A)` (attribute)

Returns: FacePoset

Computes the face poset of the regular cell complex having the homotopy type of Milnor fiber of the arrangement *A*.

Example
```
gap> A:=AGpql(2,2,3);
<HyperplaneArrangement: 6 hyperplanes in 3-space>
gap> MCpx:=MilnorFiberComplex(A);
<FacePoset of dimension 2 with f-vector [ 12, 60, 60 ]>
```

8.1.3 FPtoCWCpx

▷ `FPtoCWCpx(FP)` (function)

Returns: HAP CW Complex

Converts a HYPARR `FacePoset` into a HAP `RegularCWComplex` for further computations of topological invariants.

Example

```
gap> A:=AGpql(2,2,3);
<HyperplaneArrangement: 6 hyperplanes in 3-space>
gap> MCpx:=MilnorFiberComplex(A);
<FacePoset of dimension 2 with f-vector [ 12, 60, 60 ]>
gap> MCW := FPtoCWCpx(MCpx);
Regular CW-complex of dimension 2
gap> Homology(MCW,0);
[ 0 ]
gap> Homology(MCW,1);
[ 0, 0, 0, 0, 0, 0, 0 ]
```

Chapter 9

Tope posets

9.1 Operations

9.1.1 TopePoset (for IsOrientedMatroid, IsList)

▷ TopePoset(*OM, BTope*) (operation)

Returns: A TopePoset object

Constructs the tope poset of the oriented matroid *OM* with respect to the base tope *BTope* (given as sign vector).

Example

```
gap> O:=OM([[1,0],[0,1]]); Ts:=OMCovectors(O)[1]; T:=Ts[1];
<OrientedMatroid: 2 elements, rank 2>
[ [ -1, -1 ], [ 1, 1 ], [ -1, 1 ], [ 1, -1 ] ]
[ -1, -1 ]
gap> TP:=TopePoset(O,T);
<TopePoset with 3 topes>
```

9.2 Attributes

9.2.1 TPGroundSet (for IsTopePoset)

▷ TPGroundSet(*TP*) (attribute)

Returns: A list of lists.

The ground set of the tope poset *TP*. It is a list of list, recording the separating elements of the given tope from the base tope.

Example

```
gap> TPGroundSet(TP);
[ [ [  ] ], [ [ 1 ], [ 2 ] ], [ [ 1, 2 ] ] ]
```

9.2.2 TPBaseTope (for IsTopePoset)

▷ TPBaseTope(*TP*) (attribute)

Returns: A sign vector.

Returns the bast tope of the tope poset *TP*.

Example

```
gap> TPBaseTope(TP);
[ -1, -1 ]
```

9.2.3 TPRankPoly (for IsTopePoset)

▷ `TPRankPoly(TP)` (attribute)

Returns: A polynomial.

Returns the rank generating polynomial of the tope poset *TP*.

Example

```
gap> TPRankPoly(TP);
t^2+2*t+1
```

Chapter 10

Varchenko-Gelfand algebra

10.1 Construction

10.1.1 VGAlgebra (for IsOrientedMatroid, IsField)

▷ `VGAlgebra(`*`OM, field`*`)` (operation)

Returns: Varchenko-Gelfand albgera

Constructs the Varchenko-Gelfand algebra of *`OM`*.

Example

```
```

Chapter 11

Visualization of real 3-arrangements

11.1 Global methods

11.1.1 LaTeXDrawProjPicture

▷ `LaTeXDrawProjPicture(A[, DrawOptions])` (function)

Returns: A string.

Generates LaTeX tikz-code for a nice projective picture of the real 3-arrangement. If x, y, z are the coordinates, by default, this is the 2-dim affine arrangement obtained by intersecting *A* with the plane $z = 1$.

In the optional parameters *DrawOptions*, a record, the following parameters can be chosen

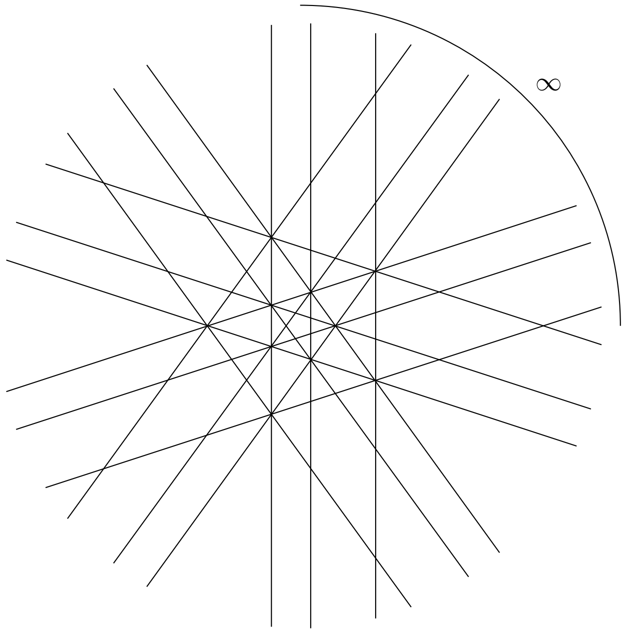
- *scale*, a positive rational scaling factor,
- *isecps* intersection points are drawn, `true` or `false`,
- *Hind* labels for the hyperplanes are added, `true` or `false`,
- *deconeH* a vector giving the normal of the plane (=1) with which to intersect *A*,
- *MarkHs* a list of indices of hyperplanes of *A*, these are drawn in another color.

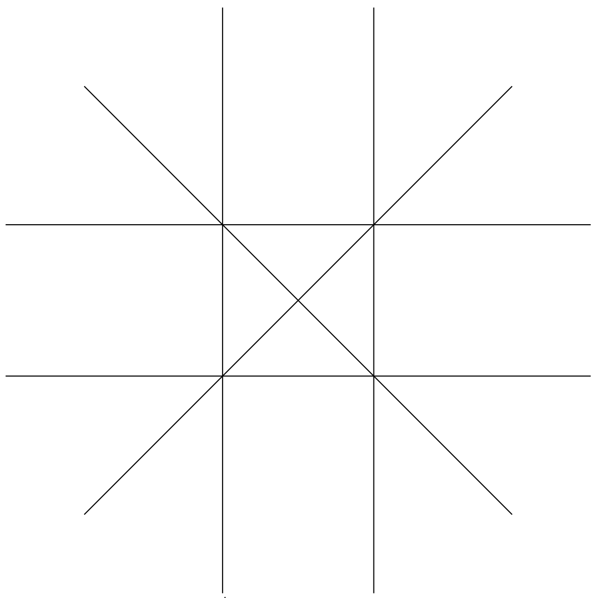
Example

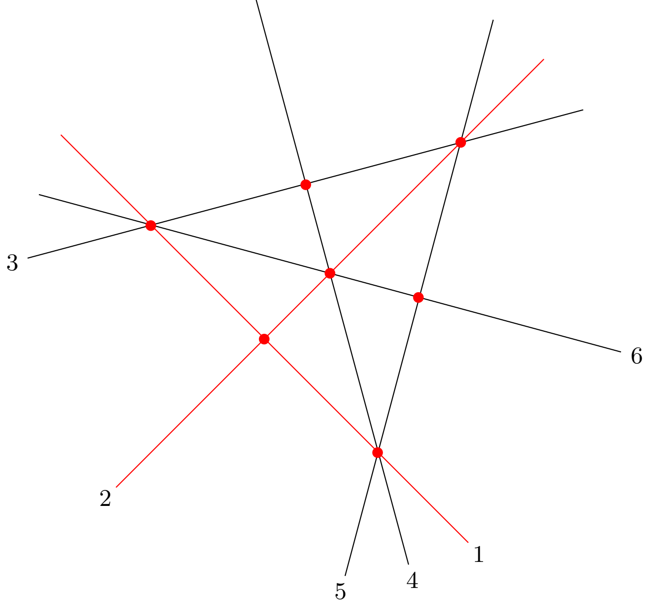
```
gap> A:=AGpql(2,2,3);
<HyperplaneArrangement: 6 hyperplanes in 3-space>
gap> Print(LaTeXDrawProjPicture(A));
\begin{tikzpicture}[scale=1.0]
\draw (-2.83,2.83) -- (2.83,-2.83);  % H_1
\draw (2.83,2.83) -- (-2.83,-2.83);  % H_2
\draw (-1.,3.87) -- (-1.,-3.87);  % H_3
\draw (1.,3.87) -- (1.,-3.87);  % H_4
\draw (3.87,-1.) -- (-3.87,-1.);  % H_5
\draw (3.87,1.) -- (-3.87,1.);  % H_6
\end{tikzpicture}
gap> DrawOpts:=rec(scale:=1/2,isecps:=true,Hind:=true,
>       deconeH:=[1,1,1],MarkHs:=[1,2]);
gap> Print(LaTeXDrawProjPicture(AGpql(2,2,3),DrawOpts));
\begin{tikzpicture}[scale=1.0]
\draw[color=red] (-3.56,1.83) -- (1.83,-3.56);  % H_1
```

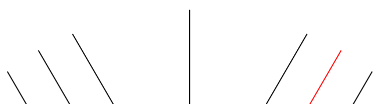
```
\node at (1.97,-3.71) {\small $1$};
\draw[color=red] (2.83,2.83) -- (-2.83,-2.83);  % H_2
\node at (-2.97,-2.97) {\small $2$};
\draw (3.35,2.16) -- (-4.,0.20);  % H_3
\node at (-4.20,0.14) {\small $3$};
\draw (-1.04,3.85) -- (1.04,-3.85);  % H_4
\node at (1.09,-4.06) {\small $4$};
\draw (2.16,3.35) -- (0.20,-4.);  % H_5
\node at (0.14,-4.20) {\small $5$};
\draw (-3.85,1.04) -- (3.85,-1.04);  % H_6
\node at (4.06,-1.09) {\small $6$};

\fill[red] (-0.87,-0.87) circle[radius=2pt];  % P[ 1, 2 ]
\fill[red] (-2.37,0.63) circle[radius=2pt];  % P[ 1, 3, 6 ]
\fill[red] (1.73,1.73) circle[radius=2pt];  % P[ 2, 3, 5 ]
\fill[red] (0.63,-2.37) circle[radius=2pt];  % P[ 1, 4, 5 ]
\fill[red] (0.0,0.0) circle[radius=2pt];  % P[ 2, 4, 6 ]
\fill[red] (-0.32,1.17) circle[radius=2pt];  % P[ 3, 4 ]
\fill[red] (1.17,-0.32) circle[radius=2pt];  % P[ 5, 6 ]
\end{tikzpicture}
```

The preceding example will look as follows


∞




3
6
2
1
5
4



11.1.2 LaTeXDrawSpherePicture

▷ `LaTeXDrawSpherePicture(A)` (function)

Returns: A string.

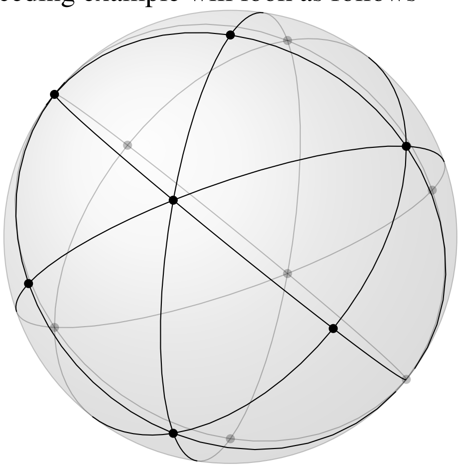
Generates LaTeX tikz-code for the intersection of a real 3-arrangement with the unit sphere. To compile the LaTeX-code the .sty-file "graphonsphere.sty" (from /doc/LaTeX_Examples) needs to be in the same folder and added via "\usepackage{graphonsphere}".

Example

```
gap> A:=AGpql(2,2,3);
<HyperplaneArrangement: 6 hyperplanes in 3-space>
gap> Print(LaTeXDrawSpherePicture(A));
\tdplotsetmaincoords{63}{63}
\begin{tikzpicture}[scale=3,tdplot_main_coords]

\draw[ball color = gray!40, opacity = 0.2] (0,0,0) circle (1cm);

\def\OMcolor{black}
{\color{\OMcolor}
% the vertices of the OM-cpx
\foreach \i/\vx/\vy/\vz in {
1/0./0./1.,
2/-0.57/0.57/0.57,
3/-0.57/-0.57/0.57,
4/0.57/-0.57/0.57,
5/0.57/0.57/0.57,
6/0./1./0.,
7/1./0./0.,
8/0./0./-1.,
9/0.57/-0.57/-0.57,
10/0.57/0.57/-0.57,
11/-0.57/0.57/-0.57,
12/-0.57/-0.57/-0.57,
13/0./-1./0.,
14/-1./0./0.
}{
\node (\i) at (\vx,\vy,\vz) {};
\POnSfb{\vx}{\vy}{\vz}{}
}
%
% the edges of the OM-cpx
\foreach \ax/\ay/\az/\bx/\by/\bz in {
0./0./1./  -0.57/0.57/0.57, %[ 1, 2 ]
0./0./1./  -0.57/-0.57/0.57, %[ 1, 3 ]
0./0./1./  0.57/-0.57/0.57, %[ 1, 4 ]
0./0./1./  0.57/0.57/0.57, %[ 1, 5 ]
-0.57/0.57/0.57/  -0.57/-0.57/0.57, %[ 2, 3 ]
-0.57/0.57/0.57/  0.57/0.57/0.57, %[ 2, 5 ]
-0.57/0.57/0.57/  0./1./0., %[ 2, 6 ]
-0.57/0.57/0.57/  -0.57/0.57/-0.57, %[ 2, 11 ]
-0.57/0.57/0.57/  -1./0./0., %[ 2, 14 ]
-0.57/-0.57/0.57/  0.57/-0.57/0.57, %[ 3, 4 ]
-0.57/-0.57/0.57/  -0.57/-0.57/-0.57, %[ 3, 12 ]
```

```
-0.57/-0.57/0.57/  0./-1./0., %[ 3, 13 ]
-0.57/-0.57/0.57/  -1./0./0., %[ 3, 14 ]
0.57/-0.57/0.57/  0.57/0.57/0.57, %[ 4, 5 ]
0.57/-0.57/0.57/  1./0./0., %[ 4, 7 ]
0.57/-0.57/0.57/  0.57/-0.57/-0.57, %[ 4, 9 ]
0.57/-0.57/0.57/  0./-1./0., %[ 4, 13 ]
0.57/0.57/0.57/  0./1./0., %[ 5, 6 ]
0.57/0.57/0.57/  1./0./0., %[ 5, 7 ]
0.57/0.57/0.57/  0.57/0.57/-0.57, %[ 5, 10 ]
0./1./0./  0.57/0.57/-0.57, %[ 6, 10 ]
0./1./0./  -0.57/0.57/-0.57, %[ 6, 11 ]
1./0./0./  0.57/-0.57/-0.57, %[ 7, 9 ]
1./0./0./  0.57/0.57/-0.57, %[ 7, 10 ]
0./0./-1./  0.57/-0.57/-0.57, %[ 8, 9 ]
0./0./-1./  0.57/0.57/-0.57, %[ 8, 10 ]
0./0./-1./  -0.57/0.57/-0.57, %[ 8, 11 ]
0./0./-1./  -0.57/-0.57/-0.57, %[ 8, 12 ]
0.57/-0.57/-0.57/  0.57/0.57/-0.57, %[ 9, 10 ]
0.57/-0.57/-0.57/  -0.57/-0.57/-0.57, %[ 9, 12 ]
0.57/-0.57/-0.57/  0./-1./0., %[ 9, 13 ]
0.57/0.57/-0.57/  -0.57/0.57/-0.57, %[ 10, 11 ]
-0.57/0.57/-0.57/  -0.57/-0.57/-0.57, %[ 11, 12 ]
-0.57/0.57/-0.57/  -1./0./0., %[ 11, 14 ]
-0.57/-0.57/-0.57/  0./-1./0., %[ 12, 13 ]
-0.57/-0.57/-0.57/  -1./0./0. %[ 12, 14 ]
}{
\GCArcABfb{\ax}{\ay}{\az}{\bx}{\by}{\bz}{color=\OMcolor}
}
}
\end{tikzpicture}
```

The preceding example will look as follows



Sphere picture produced by LaTeXDrawSpherePicture.

11.1.3 LaTeXDrawTopeGraph

▷ `LaTeXDrawTopeGraph(A)` (function)

Returns: A string.

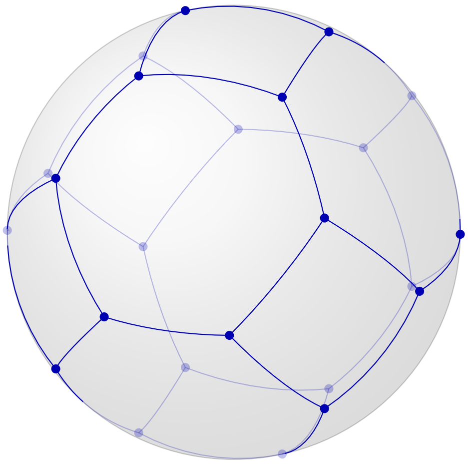
Generates LaTeX tikz-code for the tope graph of a real 3-arrangement on the unit sphere. To compile the LaTeX-code the .sty-file "graphonsphere.sty" (from /doc/LaTeX_Examples) needs to be in the same folder and added via "\usepackage{graphonsphere}".

Example

```
gap> A:=AGpql(2,2,3);
<HyperplaneArrangement: 6 hyperplanes in 3-space>
gap> Print(LaTeXDrawTopeGraph(A));
\tdplotsetmaincoords{63}{63}
\begin{tikzpicture}[scale=3,tdplot_main_coords]
\draw[ball color = gray!40, opacity = 0.2] (0,0,0) circle (1cm);
\def\OMcolor{blue!70!black}
{\color{\OMcolor}
% the vertices of the OM-cpx
\foreach \i/\vx/\vy/\vz in {
1/-0.88/-0.47/0.,
2/0.88/0.47/0.,
3/-0.88/0./-0.47,
4/0.88/0./0.47,
5/-0.88/0./0.47,
6/0.88/0./-0.47,
7/-0.88/0.47/0.,
8/0.88/-0.47/0.,
9/-0.47/0./-0.88,
10/0.47/0./0.88,
11/-0.47/0./0.88,
12/0.47/0./-0.88,
13/-0.47/-0.88/0.,
14/0.47/0.88/0.,
15/0./-0.47/-0.88,
16/0./0.47/0.88,
17/0./-0.47/0.88,
18/0./0.47/-0.88,
19/0./-0.88/-0.47,
20/0./0.88/0.47,
21/0./-0.88/0.47,
22/0./0.88/-0.47,
23/0.47/-0.88/0.,
24/-0.47/0.88/0.
}{
\node (\i) at (\vx,\vy,\vz) {};
\POnSfb{\vx}{\vy}{\vz}{}
}
%
% the edges of the OM-cpx
\foreach \ax/\ay/\az/\bx/\by/\bz in {
-0.88/-0.47/0./  -0.88/0./-0.47, %[ 1, 3 ]
-0.88/-0.47/0./  -0.88/0./0.47, %[ 1, 5 ]
-0.88/-0.47/0./  -0.47/-0.88/0., %[ 1, 13 ]
```

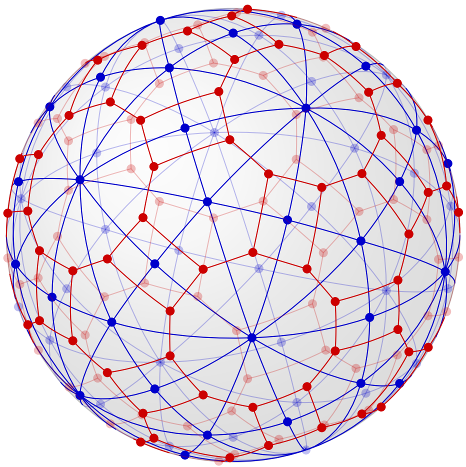
```
0.88/0.47/0./  0.88/0./0.47, %[ 2, 4 ]
0.88/0.47/0./  0.88/0./-0.47, %[ 2, 6 ]
0.88/0.47/0./  0.47/0.88/0., %[ 2, 14 ]
-0.88/0./-0.47/  -0.88/0.47/0., %[ 3, 7 ]
-0.88/0./-0.47/  -0.47/0./-0.88, %[ 3, 9 ]
0.88/0./0.47/  0.88/-0.47/0., %[ 4, 8 ]
0.88/0./0.47/  0.47/0./0.88, %[ 4, 10 ]
-0.88/0./0.47/  -0.88/0.47/0., %[ 5, 7 ]
-0.88/0./0.47/  -0.47/0./0.88, %[ 5, 11 ]
0.88/0./-0.47/  0.88/-0.47/0., %[ 6, 8 ]
0.88/0./-0.47/  0.47/0./-0.88, %[ 6, 12 ]
-0.88/0.47/0./  -0.47/0.88/0., %[ 7, 24 ]
0.88/-0.47/0./  0.47/-0.88/0., %[ 8, 23 ]
-0.47/0./-0.88/  0./-0.47/-0.88, %[ 9, 15 ]
-0.47/0./-0.88/  0./0.47/-0.88, %[ 9, 18 ]
0.47/0./0.88/  0./0.47/0.88, %[ 10, 16 ]
0.47/0./0.88/  0./-0.47/0.88, %[ 10, 17 ]
-0.47/0./0.88/  0./0.47/0.88, %[ 11, 16 ]
-0.47/0./0.88/  0./-0.47/0.88, %[ 11, 17 ]
0.47/0./-0.88/  0./-0.47/-0.88, %[ 12, 15 ]
0.47/0./-0.88/  0./0.47/-0.88, %[ 12, 18 ]
-0.47/-0.88/0./  0./-0.88/-0.47, %[ 13, 19 ]
-0.47/-0.88/0./  0./-0.88/0.47, %[ 13, 21 ]
0.47/0.88/0./  0./0.88/0.47, %[ 14, 20 ]
0.47/0.88/0./  0./0.88/-0.47, %[ 14, 22 ]
0./-0.47/-0.88/  0./-0.88/-0.47, %[ 15, 19 ]
0./0.47/0.88/  0./0.88/0.47, %[ 16, 20 ]
0./-0.47/0.88/  0./-0.88/0.47, %[ 17, 21 ]
0./0.47/-0.88/  0./0.88/-0.47, %[ 18, 22 ]
0./-0.88/-0.47/  0.47/-0.88/0., %[ 19, 23 ]
0./0.88/0.47/  -0.47/0.88/0., %[ 20, 24 ]
0./-0.88/0.47/  0.47/-0.88/0., %[ 21, 23 ]
0./0.88/-0.47/  -0.47/0.88/0. %[ 22, 24 ]
}{
\GCArcABfb{\ax}{\ay}{\az}{\bx}{\by}{\bz}{color=\OMcolor}
}
}
\end{tikzpicture}
```

The preceding example will look as follows



Tope graph picture produced by `LaTeXDrawTopeGraph`.

Both functions `LaTexDrawSpherePicture` and `LaTeXDrawTopeGraph` combined can be used to draw nice pictures of oriented matroid complexes and their duals.



Oriented matroid complex and its dual of $\mathcal{A}(H_3)$.

Chapter 12

Realization spaces of geometric lattices

The computation of the realization space uses the SINGULAR package to call functions from SINGULAR [DGPS24]. Our implementation is an adaptation of the algorithm described in [Cun22, Algorithm 3.9] for geometric lattices of arbitrary rank.

12.1 Construction

12.1.1 LRealizationSpace (for IsGeomLattice, IsInt)

▷ LRealizationSpace(*L*, *char*[, *GenSetL*]) (operation)

Returns: RealizationSpaceOfGeometricLattice

Computes the realization space of the geometric lattice *L* in characteristic *char*. Optionally, a generating set of *L* (as a subset of `LAtoms(L)`) can be given. Otherwise, the algorithm tries to find a small one.

Example

```
gap> L:=IntersectionLattice(AGpql(3,3,3));
<Geometric lattice: 9 atoms, rank 3>
gap> RS:=LRealizationSpace(L,0);
<RealizationSpaceOfGeomLattice: in characteristic 0, non-empty: true>
gap> S:=LGenSet(L);
[ 2, 3, 4, 6, 8 ]
gap> RS:=LRealizationSpace(L,0,S);
<RealizationSpaceOfGeomLattice: in characteristic 0, non-empty: true
```

12.2 Attributes

12.2.1 LGenSet (for IsGeomLattice)

▷ LGenSet(*L*) (attribute)

Returns: a set

Computes some generating set of *L*, see [Cun22, Def.3.4] resp. [GMRS26, Def.4.2].

Example

```
gap> L:=IntersectionLattice(AGpql(3,3,3));
<Geometric lattice: 9 atoms, rank 3>
gap> LGenSet(L);
[ 2, 3, 4, 6, 8 ]
```

12.2.2 RSLattice (for IsRealizationSpaceOfGeomLattice)

▷ `RSLattice(RS)` (attribute)

Returns: a geometric lattice

Returns the geometric lattice of the realization space *RS*.

Example

```
gap> RSLattice(RS);
<Geometric lattice: 9 atoms, rank 3>
```

12.2.3 RSCharacteristic (for IsRealizationSpaceOfGeomLattice)

▷ `RSCharacteristic(RS)` (attribute)

Returns: 0 or a prime number

Returns the characteristic of the reliation space *RS*.

Example

```
gap> RSCharacteristic(RS);
0
```

12.2.4 RSDefField (for IsRealizationSpaceOfGeomLattice)

▷ `RSDefField(RS)` (attribute)

Returns: a field

Returns the field over which the realization space *RS* is defined.

Example

```
gap> RSDefField(RS);
Rationals
```

12.2.5 RSCoeffMat (for IsRealizationSpaceOfGeomLattice)

▷ `RSCoeffMat(RS)` (attribute)

Returns: a matrix

Returns the coefficient matrix of *RS* with entries in `RSPRing(RS)`.

Example

```
gap> RSCoeffMat(RS);
[ [ a1-1, a1, 0 ], [ 1, 0, 0 ], [ 0, 1, 0 ], [ 0, 0, 1 ],
  [ 1, 1, a1 ], [ 1, 1, 1 ], [ 0, 1, -a1^2+a1 ],
  [ 1, 0, a1 ], [ -a1+1, -a1, -a1^2 ] ]
```

12.2.6 RSDimension (for IsRealizationSpaceOfGeomLattice)

▷ `RSDimension(RS)` (attribute)

Returns: a non-negative integer

Returns the dimension of *RS*.

Example

```
gap> RSDimension(RS);
0
```

12.2.7 RSPRing (for IsRealizationSpaceOfGeomLattice)

▷ `RSPRing(RS)` (attribute)

Returns: a polynomial ring

The polynomial ring over which the representing coefficient matrix `RSCoeffMat(RS)` of *RS* is defined.

Example
```
gap> RSPRing(RS);
Rationals[a1]
```

12.2.8 RSIdealMinors (for IsRealizationSpaceOfGeomLattice)

▷ `RSIdealMinors(RS)` (attribute)

Returns: an ideal

Returns the ideal of equations which have to satisfied due to the dependcy of atoms in `RSLattice(RS)` over `RSDefField(RS)`.

Example
```
gap> I:=RSIdealMinors(RS); GeneratorsOfIdeal(I);
<two-sided ideal in Rationals[a1], (1 generator)>
[ a1^2-a1+1 ]
```

12.2.9 RSNonMinors (for IsRealizationSpaceOfGeomLattice)

▷ `RSNonMinors(RS)` (attribute)

Returns: a list of polynomials

Gives a list of polynomials which should not vanish due to independent subsets of atoms in `RSLattice(RS)` over `RSDefField(RS)`.

Example
```
gap> RSNonMinors(RS);
[ a1^3-a1^2, a1^3, -a1^3+2*a1^2-a1, -2*a1^2+2*a1-1 ]
```

12.2.10 RSEvalPoint (for IsRealizationSpaceOfGeomLattice)

▷ `RSEvalPoint(RS)` (attribute)

12.3 Properties

12.3.1 RSIsNonEmpty (for IsRealizationSpaceOfGeomLattice)

▷ `RSIsNonEmpty(RS)` (property)

Returns: `true` or `false`

12.4 Further (auxillary) Operations

12.4.1 LSubsetGeneratedByS (for IsGeomLattice,IsList)

▷ `LSubsetGeneratedByS(L, S)` (operation)

12.4.2 LIsGenSet (for IsGeomLattice,IsList)

▷ `LIsGenSet(L, S)` (operation)

Chapter 13

Greedy search for arrangements

This is an implementation of a basic greedy algorithm to construct arrangements satisfying a certain property, based on [Cun22]. A variant was used in [MMR24] to construct an arrangement which is 4-formal with a restriction which is not 3-formal.

13.1 Constructions

13.1.1 HArrGreedySearch (for IsInt,IsInt,IsField,IsFunction,IsInt, IsRat)

▷ `HArrGreedySearch(NumberOfHs, dim, GField, PropTargetFct, MaxNoIterations, heat)` (operation)

Returns: A hyperplane arrangement

Constructs a greedy search for arrangements over *GField* with *NumberOfHs* hyperplanes in dimension *dim*.

- *PropTargetFct* should be a function taking a hyperplane arragement as argument, returning a non-negative real number, which the search tries to minimize, and such that value=0 means the arrangement satisfies the desired property.
- *MaxNoIterations* is the maximal number of steps to be carried out in the search (one step = exchanging a hyperplane).
- *heat* is a margin for the target function

Example

```
gap> HArrGreedySearch(13,3,GF(17),CharPolySplits,400);
GreedySearch over GF(17) for arrangements:
 - of rank 3
 - with 13 hyperplanes.
```

13.1.2 CharPolySplits

▷ `CharPolySplits(A)` (function)

A target/score function for splitting of the characteristic polynomial of *A* over $\mathbb{Z}$.

13.1.3 DistMSetInvL

▷ `DistMSetInvL(A, B)` (function)

A target/score function for comparing `MSetInvL(B)` and `MSetInvL(B)` (see `MSetInvL` (2.2.7)).

13.2 Attributes

13.2.1 GreedySearchRun (for IsHArrGreedySearch)

▷ `GreedySearchRun(GS)` (attribute)

Returns: a function

Run the search algorithm of *GS* which will return an arrangement or `fail`.

Example

```
gap> GS:=HArrGreedySearch(13,3,GF(17),CharPolySplits,400);
GreedySearch over GF(17) for arrangements:
 - of rank 3
 - with 13 hyperplanes.
gap> A:=GreedySearchRun(GS)();
227 Iterations - <HyperplaneArrangement: 13 hyperplanes in 3-space>
gap> ExpArr(A);
[ 1, 6, 6 ]
gap> HArrIsSupersolvable(A);
true
gap> A:=GreedySearchRun(GS)();
358 Iterations - <HyperplaneArrangement: 13 hyperplanes in 3-space>
gap> ExpArr(A);
[ 1, 6, 6 ]
gap> HArrIsSupersolvable(A);
false
gap> HArrIsFree(A);
true
```

13.2.2 GreedySearchGF (for IsHArrGreedySearch)

▷ `GreedySearchGF(GS)` (attribute)

Returns: a field

The Galois field over which to search for arrangements.

Example

```
```

13.2.3 GreedySearchDimension (for IsHArrGreedySearch)

▷ `GreedySearchDimension(GS)` (attribute)

Example

```
```

13.2.4 GreedySearchNOfHs (for IsHArrGreedySearch)

▷ `GreedySearchNOfHs(GS)` (attribute)

Example

```
```

13.2.5 GreedySearchTargetFct (for IsHArrGreedySearch)

▷ `GreedySearchTargetFct(GS)` (attribute)

Example

```
```

13.2.6 GreedySearchMaxIterations (for IsHArrGreedySearch)

▷ `GreedySearchMaxIterations(GS)` (attribute)

Example

```
```

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